

Thuan Doan

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WORK EXPERIENCE

HCLTech

Machine Learning Engineer / Researcher

Ho Chi Minh, Viet Nam

Jun 2023 – Present

- Developed and fine-tuned domain-specific open LLMs for automation software testing at ANZ bank.
- Designed and implemented a RAG-based chatbot, integrated with internal tools and domain knowledge.
- Researched and applied cutting-edge LLM techniques: finetuning, RLHF, optimization, RAG techniques, knowledge graph, prompt engineering, planning and reasoning, multi-agent system, code completion.
- Applied best practices for data management, model training, evaluation, deployment and monitoring.

Infinigru

Machine Learning Engineer

Ho Chi Minh, Viet Nam

Aug 2018 – May 2023

- Developed ML algorithms and applied AI techniques in various domains including vision, NLP, fintech.
- Conducted data analytics, feature engineering, model tuning and evaluation to optimize solution.
- Built and maintained end-to-end ML pipelines, automating data preprocessing, model training and deployment using cloud platforms. Monitored and retrained models as needed.

Global Outsource Solution

Technical Lead

Ho Chi Minh, Viet Nam

Mar 2015 – Jun 2018

- Led a cross-functional team of software engineers, designers, and QA specialists.
- Architected and led the development of scalable software systems, employing microservices, domain-driven design, and event-driven architectures to meet evolving business needs.
- Applied CI/CD and maintained coding standards, high code quality within the team.

Forix

Software Engineer

Ho Chi Minh, Viet Nam

Mar 2012 – Mar 2014

- Developed web applications in various domains including e-commerce, social media, data-warehouse, social media, education.
- Designed RESTful APIs to support mobile and front-end systems.

Global CyberSoft

Web Developer

Ho Chi Minh, Viet Nam

Sep 2008 – Sep 2010

- Designed and developed web applications using PHP frameworks: Zend, Drupal, WordPress, Joomla.

SKILLS

- Programming Languages: Python, Java, PHP, JavaScript, .Net, C++.
- AI Tools: TensorFlow, PyTorch, Scikit-learn, Hugging Face, Pandas, Numpy, Matplotlib, OpenCV, NLTK.
- ML Algorithms: Regression, Classification, SVM, Tree-based Models, Deep Learning, CNN, RNN, Transformers, Deep Generative Models, Large Language Models.
- Backend Frameworks: FastAPI, Spring, Laravel, Flask, Magento, WordPress.
- Frontend Frameworks: NodeJS, React, VueJS, Angular, jQuery.
- Databases: MySQL, PostgreSQL, MongoDB, Cassandra, Redis.
- Tools: Docker, Kubernetes, Jupyter Notebooks, Git, AWS, GCP

EDUCATION

University of Information Technology

Ho Chi Minh, Viet Nam

2020 – 2023

- Master's Degree in Computer Science.
- Researcher Assistant at MMLab. Topics: knowledge engineering, object segmentation, depth estimation, deep generative models, NLP, Large Language Models (LLMs).

Japan Advanced Institute of Science and Technology (JAIST)

Ishikawa, Japan

2020

- Researcher Assistant under supervision of Prof. Kazuhiro Ogata. Topic: Theorem Proving.

Kookmin University

Seoul, Korea

2011– 2012

- Researcher Assistant under supervision of Prof. Young Kim. Topic: Security with SELinux.

University of Science

Ho Chi Minh, Viet Nam

2004 – 2008

- Bachelor's Degree in Computer Science. Major in software engineering.

Certifications:

- TensorFlow Developer [Certificate](#).
- Coursera specializations about Math, Machine Learning, Deep Learning, Vision, NLP, Data Science ([link](#))
- IELTS: 6.5

Publication:

- Efficient Finetuning Large Language Models For Vietnamese Chatbot (first author) - [MAPR 2023](#). [Paper's link](#)

PROJECTS

EnsayoAI | Transformers, LangGraph, vLLM, OpenAI, langfuse, nemo guardrails, AWS Sagemaker

- Leveraged open-LLMs (Mistral, Llama 3) and applied domain-adaptive finetuning (LoRA, RLHF) to build Gen-AI powered tool for automation software testing.
- Build RAG based chatbot system. Enhanced with advanced techniques: hybrid search, semantic router, chunking, re-ranker, multi-retriever, knowledge graph, multimodal, tool-integrated agent.
- Built LLM multi-agent framework for autonomous business workflows at bank.
- Researched and applied latest techniques for AI-assistant software development: retrieval augmented code generation, self-reflection, reasoning & planning, agentic patterns.

Customer Support Assistant | audio processing, wav2vec, Kaldi, Rasa, NLTK, SpaCy

- Optimized BERT-based models for Named Entity Recognition (NER) and Sentiment Analysis
- Developed and fine-tuned speech recognition models in Korean and Vietnamese
- Built a rule-based chatbot, designed dialogue flows and response patterns for customer inquiries.

Fraud Detection System | Xgboost, Catboost, SHAP, Apache Spark/Flink, Kafka, Redis, MongoDB.

- Data analytic & model development for fraud detections using tree-based models.
- Built scalable system for real-time data processing using cloud and big data technologies.

UAV-Based Surveillance | object detection, segmentation, depth estimation, Yolo, Unet, SLAM

- Developed models for real-time detection and monitoring of wildfires and road surface defects.
- Optimized models and deployed to edge devices with Jetson Nano.